The core defines interfaces; adapters are swapped at start-up

Level 03 - ports and adapters. One port, three implementations: memory for unit tests, local or LocalStack for development, AWS or the real service in Beta.

Port (ports/, a Protocol)	Unit tests	Laptop and Alpha	Beta and external
SceneSource *find_scenes, load_stack*	in-memory arrays	local GeoTIFF folder; Sentinel-2 STAC (pystac-client)	Sentinel-2 STAC and AWS open data; scene cache in S3
ObjectStore *put, get, exists, list*	dict in memory	local disk; LocalStack S3	Amazon S3 (data lake bucket)
DetectionRepository *save, get, query by region and date*	dict in memory	LocalStack DynamoDB	DynamoDB on demand (D12)
ReviewRepository *append review, history*	list in memory	LocalStack DynamoDB	DynamoDB on demand
RegionRepository *save, list, alert cadence*	dict in memory	LocalStack DynamoDB	DynamoDB on demand
WindProvider *wind_at(point, time) with source and version*	constant or recorded wind	ERA5 through the CDS API, cached files	ERA5 and ERA5T; final values replace preliminary ones
FacilityIndex *near(point, radius)*	small list in memory	GeoJSON or Parquet of public facility data	same file, versioned in the data lake
EvidenceStore *write once, read, verify*	dict, write once	read-only bucket in LocalStack	S3 Object Lock, WORM (NFR2)
Signer *sign(manifest), verify*	fixed test key	Ed25519 key from an untracked file	key custody is a Beta decision (ADR)
AuthProvider *authenticate(token), user and role*	fixed users	JWT with argon2 password hashes	Cognito (planned)
AuditSink *append(entry), verify chain*	list in memory	LocalStack DynamoDB	DynamoDB, immutable (NFR3)
ModelRegistry *register, approved_version, load*	dict in memory	local MLflow server	MLflow and S3 artefacts

Rule from the report: do not buy services early. Alpha runs entirely on the first two columns (credit is $200, decision D13); the third column is scaffolded as Terraform and adapters in Beta. A port exists only if something in services/ needs it; a new port needs an ADR line explaining which test or adapter it makes possible.